
Air Quality Index Prediction System

Project Id: AIML-15

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Problem Description(Overview for AQI):-

The Air Quality Index (AQI) Prediction System is a Python-based machine learning application developed to analyze environmental data and predict air quality levels. Air pollution has become a major concern affecting human health and the environment. This system automates the process of collecting, preprocessing, and analyzing air pollutant data to predict AQI values and classify them into categories such as Good, Moderate, and Unhealthy. Using core Machine Learning concepts such as data preprocessing, feature engineering, classification algorithms, model evaluation, and visualization techniques, this project minimizes manual effort and provides accurate insights into air quality conditions. It helps in understanding pollution trends and supports decision-making for environmental monitoring.

Architecture:

Flow of the system:

- **Data Collection**

Dataset containing pollutant values (PM2.5, PM10, CO, NO2, O3, etc.)

- **Data Preprocessing**

1. Handling missing values
2. Removing outliers using IQR
3. Converting negative values to valid values
4. Renaming columns for clarity

- **Feature Engineering**

Creation of AQI categories (Good, Moderate, Unhealthy, etc.)

- **Feature Scaling**

StandardScaler / MinMaxScaler applied

- **Model Training**

1. Logistic Regression
2. K-Nearest Neighbors (KNN)
3. Support Vector Machine (SVM)
4. Ridge and Lasso Regression

- **Model Evaluation**

Accuracy comparison of models

- **Model Saving**

Best model stored using pickle



Fig.1 Sample flow diagram.

Models Used(Applied/ Planned) :-

1. Logistic Regression

A supervised learning algorithm used for classification. It predicts AQI categories based on input features and provides good baseline accuracy.

2. K-Nearest Neighbors (KNN)

A distance-based algorithm that classifies AQI categories based on similarity with nearest data points.

3. Support Vector Machine (SVM)

An advanced classification algorithm that finds the optimal boundary between AQI categories. It provided the best performance in this project.

4. Ridge Regression

A regularization technique used to handle multicollinearity and improve model generalization.

5. Lasso Regression

Used for feature selection by reducing less important feature coefficients to zero.

Evaluation metrics used

The following metrics were used to evaluate model performance:

- **Accuracy Score:**
 - Measures the percentage of correct predictions made by the model.
- **Confusion Matrix**
 - Shows correct and incorrect classifications for each AQI category.

Model Comparison

- Accuracy of Logistic Regression, KNN, and SVM compared.
- **Feature Importance Analysis**
 - Using Lasso and Ridge to determine impactful features.

5. Preliminary Results and Visualizations

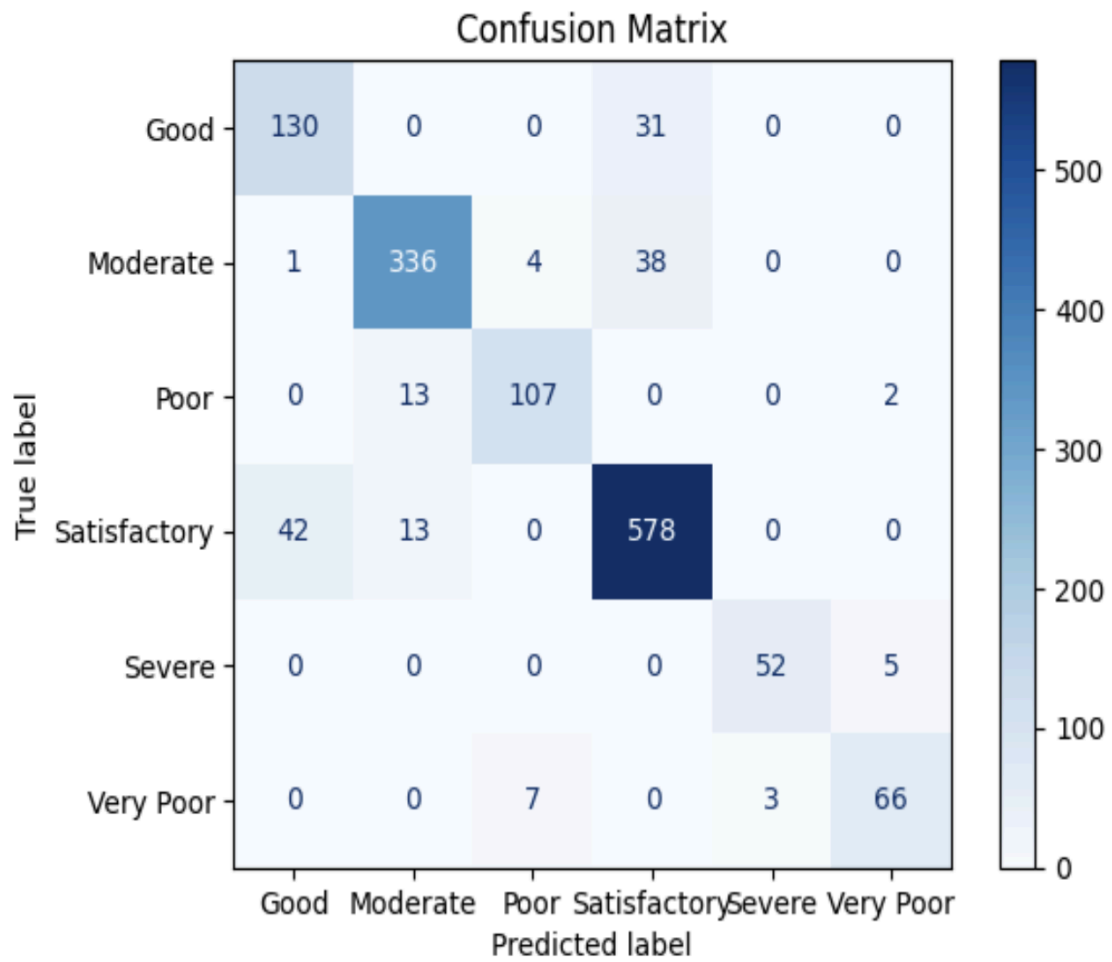
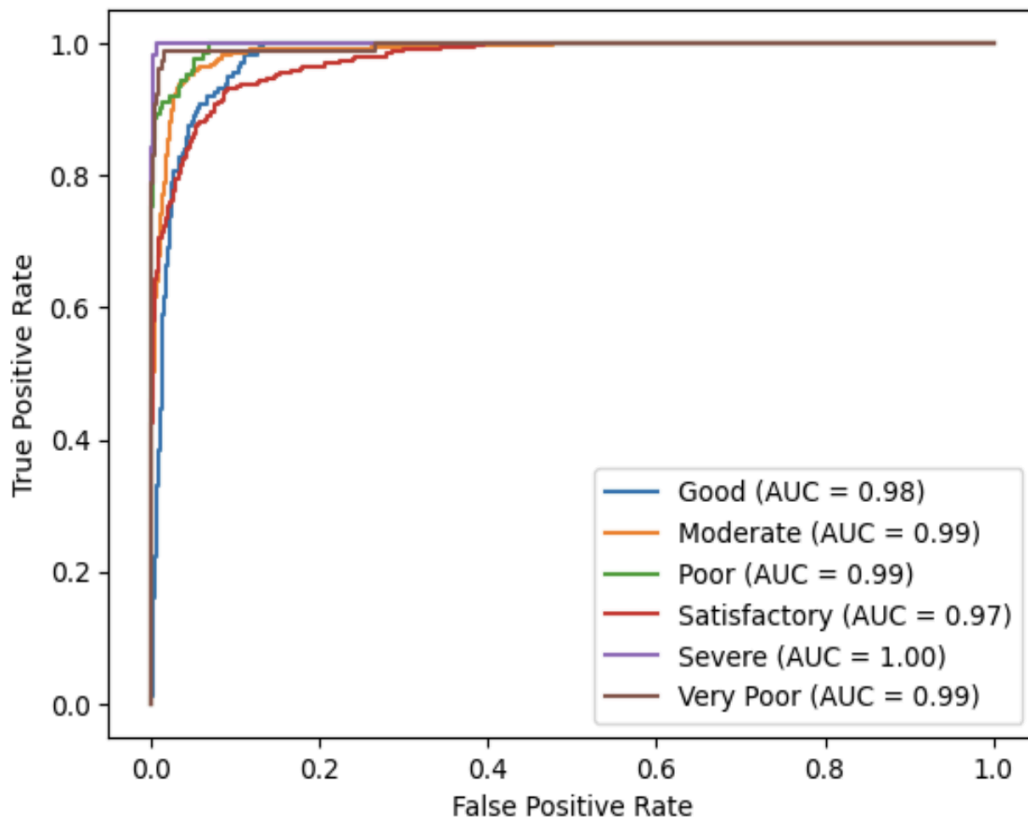


Fig.2 Confusion Matrix



ROC Curve

	precision	recall	f1-score	support
0	1.00	1.00	1.00	1
1	1.00	1.00	1.00	1
accuracy			1.00	2
macro avg	1.00	1.00	1.00	2
weighted avg	1.00	1.00	1.00	2

6. References:

- Books:
 - Let Us Python – Yashavant Kanetkar
- Online Resources:
 - GeeksforGeeks – <https://www.geeksforgeeks.org/>
 - TutorialsPoint – <https://www.tutorialspoint.com/cprogramming/index.htm>
 - Stack Overflow – <https://stackoverflow.com>

